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The AI fell short in several key areas. First, it introduced logical errors in boundary definitions, such as defining an equivalence partition for product quantity as > 1 instead of the specified ≥ 1 , thereby excluding the valid edge case of 1. Second, it violated the single-fault assumption by compounding multiple variables—like expired status and below-threshold values—into a single test case. Additionally, the AI hallucinated wrong API endpoints (e.g., `GET /api/admin/coupons` instead of `GET /api/coupons`), created duplicate test cases (BVA OFF points overlapping DT invalid representatives). The AI failed to catch these issues because it operates on local context window heuristics and pattern recognition rather than strict logical verification. Lacking real-time cross-referencing capabilities, it assumed standard RESTful routes or generic boundary styles instead of strictly aligning with the specific project requirements. Furthermore, it struggled with structural dependencies, leading it to output high-level summaries while neglecting to generate the detailed, individual test files. Ultimately, collaborating on this assignment highlighted the critical “trust but verify” principle. AI tools are exceptional at brainstorming and drafting test scaffolds, but they lack the rigorous semantic understanding required for precision engineering. Consequently, human oversight is essential to audit equivalence partitions, ensure strict adherence to single-fault assumptions, and verify that all generated test inputs match the actual system specifications.
